

Uddalok Sarkar

Indian Statistical Institute, Kolkata

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EDUCATION

- **Indian Statistical Institute** 2022 –
Ph.D., Computer Science
Advisors: Prof. Sourav Chakraborty, Prof. Kuldeep Meel
Supported by *Google PhD Fellowship*
- **Indian Statistical Institute** 2020 – 2022
M.Tech., Computer Science, Percentile: 88.24%
- **Jadavpur University** 2015 – 2019
B.E., Electrical Engineering, CGPA: 8.72/10
- **Jalpaiguri Zilla School** 2006 – 2015

RESEARCH INTERESTS

Property Testing
Verification
Model Counting

THESIS & PUBLICATIONS

- **Conference**
 1. *Instance Dependent Testing of Samplers using Interval Conditioning* - Joint work with Rishiraj Bhattacharyya, Sourav Chakraborty, Yash Pote, Sayantan Sen. (*Accepted in AAI (2024)*)
 2. *Efficient Volume Computation for SMT Formulas* - Joint work with Arijit Shaw, Kuldeep S. Meel. (*Accepted in KR (2025)*)
 3. *Assessing the Quality of Binomial Samplers: A Statistical Distance Framework* - Joint work with Sourav Chakraborty, Kuldeep S. Meel. (*Accepted in CAV (2025)*)
 4. *Testing Self Reducible Samplers* - Joint work with Rishiraj Bhattacharyya, Sourav Chakraborty, Yash Pote, Sayantan Sen. (*Accepted in AAI (2024)*)
 5. *Testing of Horn Samplers* - Joint work with Ansuman Banerjee, Shayak Chakraborty, Sourav Chakraborty, Kuldeep S. Meel, Sayantan Sen. (*Accepted in AISTATS (2023)*)
- **Journal**
 1. *Engineering an Efficient Approximate DNF-Counter* - Joint work with Mate Soos, Divesh Aggarwal, Sourav Chakraborty, Kuldeep S. Meel, Maciej Obremski. (*To Appear in JAIR*)
- **Others**
 1. *Zero-Shot Attribution for Large Language Models: A Distribution Testing Approach* - Joint work with Clément Canonne, Yash Pote. (*Under Submission*)
- **Thesis**
 1. Masters Thesis: *On Testing of Horn Samplers*, Indian Statistical Institute, 2022.

AWARDS & HONOURS

- *Best Student Paper Award* at 22nd International Conference on Principles of Knowledge Representation and Reasoning (2025).
- *Felicitaton for outstanding achievements* from Indian Statistical Institute, Kolkata (2024).
- *Google PhD Fellowship Award (2023)*.
- *Gold Medal for outstanding performance in M.Tech (CS) Programme (2020-22)* from Indian Statistical Institute, Kolkata.

TEACHING EXPERIENCE

- **Indian Statistical Institute**
 - Probability and Stochastic Processes 2023
 - Computing Laboratory (Teaching assistant) 2022-23
 - Specification and Verification of Programs (Teaching assistant) 2022
- **ACM Summer School**
 - Algorithms for Data Science 2023

REVIEWER EXPERIENCE

- Conference: CAV (2023), ICCAD (2025), AAI (2026), COLM (2026)

TECHNICAL SKILLS	Languages & frameworks Python, Tensorflow, Pytorch, Qiskit, CirQ, TensorFlow Quantum (TFQ) C, C++ NuSMV MATLAB, Simulink
REFERENCES	Prof. Sourav Chakraborty Advanced Computing and Microelectronics Unit (ACMU), Indian Statistical Institute mail id: sourav@isical.ac.in Prof. Kuldeep Meel School of Computer Science, Georgia Institute of Technology mail id: meel@cs.toronto.edu Prof. Arijit Ghosh Advanced Computing and Microelectronics Unit (ACMU), Indian Statistical Institute mail id: arijititkgpster@gmail.com
PAST RESEARCH EXPERIENCE	Digital Audio Processing Lab Indian Institute of Technology Bombay, India ▪ Research Intern (May – Jul 2018) • Project: Structural Segmentation of Hindustani Dhrupad Vocal Concerts • Supervisor: Prof. Preeti Rao, Electrical Engineering Department, IIT Bombay • Focus: Music Information Retrieval, Audio Signal Processing. C.V. Raman School of Physics and Music Jadavpur University, Kolkata, India ▪ Undergraduate Research Student • Supervisor: Dr. Dipak Ghosh, Professor Emeritus, Jadavpur University • Focus: Cognitive Neuroscience, Speech Signal Analysis, Music Information Retrieval.
INTERESTS	Music I am currently studying Jazz and Progressive Rock music. I am also interested in Folk music. <i>I have recently directed music of a folk short film Pathshala - The Inception of Pursuit.</i> Photography, Filming.